

Training and Evaluation of Machine Learning Models

Week 6, Day 1

Lumumba W. Victor
Trainer — Data Analyst — ML Expert

MediaCrest Training College

July 21, 2026

Abstract

This session bridges the gap between raw clinical data and robust machine learning models. We transition from exploratory analysis to building leak-proof preprocessing pipelines, training diverse classification algorithms, and establishing a rigorous baseline evaluation for predicting Atherosclerotic Heart Disease (AHD).

- **Pipelines:** Robust numerical and categorical preprocessing to prevent data leakage.
- **Model Selection:** Training 5 diverse classification algorithms to capture various data patterns.
- **Tuning:** RandomizedSearchCV with stratified cross-validation for efficient optimization.
- **Evaluation:** Comprehensive metrics on an untouched hold-out test set.

Our ultimate goal is to establish a reliable, high-performing baseline for early detection and risk stratification.

Session Objectives

- Formulate the clinical problem and conduct targeted Exploratory Data Analysis (EDA).
- Build robust, leak-proof preprocessing pipelines using the ColumnTransformer concept.
- Handle numerical and categorical data appropriately to ensure data integrity.
- Understand and train multiple classification algorithms (LR, KNN, SVM, DT, RF).
- Master the concepts behind hyperparameter tuning and cross-validation.
- Evaluate baseline models on an untouched hold-out test set using a comprehensive suite of clinical metrics.
- Interpret feature importance to extract clinical insights.

Agenda

- ➊ **Problem Formulation & EDA** (Dataset overview, missing values, target distribution)
- ➋ **Building Robust Preprocessing Pipelines** (Numerical, Categorical, ColumnTransformer)
- ➌ **Model Selection & Hyperparameter Tuning** (Algorithms, Grids, RandomizedSearchCV)
- ➍ **Baseline Evaluation on Hold-Out Test Set** (Metrics, Confusion Matrices, ROC Curves)
- ➎ **Feature Importance & Clinical Interpretation** (Gini vs. Permutation importance)

Part 1: Problem Formulation & Exploratory Data Analysis

"Understanding the data before modeling."

The Core Mission

Predict the Presence or Absence of Atherosclerotic Heart Disease (AHD) to enable early detection and risk stratification.

- **Early Detection:** Identifying at-risk patients before severe cardiac events occur.
- **Risk Stratification:** Allowing clinicians to prioritize interventions based on predicted probability.
- **Data-Driven Decisions:** Moving from subjective assessments to objective, data-backed diagnostic support.

Dataset Overview

- **Source:** Clinical dataset (Heart.csv) containing patient records.
- **Size:** 303 observations (patients) and 15 variables.
- **Feature Types:** A mix of demographic, vital sign, and diagnostic test results.
- **Target Variable:** AHD (Atherosclerotic Heart Disease), mapped to binary classes (0 = No, 1 = Yes).

Data Integrity

The dataset is relatively small but rich in clinical signals, requiring careful handling to avoid overfitting and ensure generalization.

Target Variable Distribution

- **No AHD (Class 0):** 164 patients (54.1%)
- **Yes AHD (Class 1):** 139 patients (45.9%)

Class Balance Analysis

The dataset is slightly imbalanced but generally well-distributed.

- Standard accuracy is a viable metric, but it must be contextualized with Precision, Recall, and ROC-AUC.
- Stratified sampling is essential during data partitioning to maintain this ratio in both training and testing sets.

Missing Values Analysis

- Most variables are complete, ensuring strong baseline signals.
- **Ca (Number of major vessels colored)**: 4 missing values.
- **Thal (Thalassemia test result)**: 2 missing values.

The Imputation Imperative

Dropping these rows would result in a negligible loss of data, but implementing robust imputation strategies within our pipeline ensures the model can handle missing data in future, real-world deployments without crashing.

Feature Categorization

To build an effective preprocessing pipeline, features must be grouped by their data type and required transformations.

Numeric Features:

- Age
- RestBP (Resting Blood Pressure)
- Chol (Cholesterol)
- MaxHR (Maximum Heart Rate)
- Oldpeak (ST depression)
- Ca (Vessels colored)

Categorical Features:

- Sex
- ChestPain
- Fbs (Fasting Blood Sugar)
- RestECG (Resting ECG)
- ExAng (Exercise Angina)
- Slope
- Thal

Part 2: Building Robust Preprocessing Pipelines

"Ensuring data integrity and preventing leakage."

Numerical Pipeline Strategy

- **Imputation:** Median Imputation.
 - Clinical data (e.g., Cholesterol, Blood Pressure) often contains extreme outliers.
 - The median is robust against outliers, providing a more representative central tendency than the mean.
- **Scaling:** StandardScaler (Z-score normalization).
 - Transforms features to have a mean of 0 and standard deviation of 1.
 - Crucial for distance-based algorithms (KNN, SVM) and gradient-based optimizers (Logistic Regression).

Categorical Pipeline Strategy

- **Imputation:** Most Frequent (Mode) Imputation.
 - Replaces missing values (e.g., in `Ca` and `Thal`) with the most common category observed in the training data.
- **Encoding:** `OrdinalEncoder`.
 - Converts categorical text labels into numerical representations.
 - **Crucial Configuration:** Set to handle unknown values (`unknown_value = -1`). This prevents the model from crashing when deployed in the real world and encountering unseen categories.

The ColumnTransformer: Unifying the Workflow

The Orchestration Challenge

A single dataset contains both numerical and categorical columns, each requiring a completely different preprocessing sequence.

- **The Solution:** The ColumnTransformer acts as a router.
- **How it Works:** It applies the numerical pipeline to specified numerical columns and the categorical pipeline to specified categorical columns simultaneously.
- **Leak-Proofing:** By embedding this inside a Pipeline, we ensure that imputation medians and scaling factors are calculated *only* on the training folds during cross-validation.

Data Partitioning Strategy

- **Split Ratio:** 80% Training, 20% Hold-Out Testing.
- **Stratification:** The split is stratified based on the target variable (AHD).

Why Stratify?

- Ensures that both the training and testing sets maintain the exact same class distribution (approx. 54.1% No, 45.9% Yes).
- Prevents scenarios where the test set might accidentally contain too few positive cases, leading to unreliable evaluation metrics.

Part 3: Model Selection & Hyperparameter Tuning

"Choosing the right algorithm and optimizing its configuration."

Selected Algorithms

We trained five diverse algorithms to leverage the "No Free Lunch" theorem:

- **Logistic Regression (LR):** Linear baseline, highly interpretable.
- **K-Nearest Neighbors (KNN):** Non-parametric, distance-based.
- **Support Vector Machines (SVM):** Margin-based, effective with kernel tricks.
- **Decision Tree (DT):** Rule-based, highly interpretable, prone to overfitting.
- **Random Forest (RF):** Ensemble bagging method, reduces variance, captures non-linearities.

Hyperparameter Tuning Setup

- **Search Strategy:** RandomizedSearchCV.
 - Samples 30 random combinations from the parameter grids.
 - More computationally efficient than exhaustive Grid Search while still finding near-optimal configurations.
- **Cross-Validation:** Stratified 5-Fold CV.
 - Ensures robust performance estimation across different data subsets.
- **Optimization Metric:** ROC-AUC.
 - Threshold-independent metric, ideal for evaluating overall discriminative ability in clinical tasks.

Cross-Validation Results Summary

Best Mean CV ROC-AUC scores across the 5 folds:

- **Logistic Regression:** 0.8996
- **SVM:** 0.8982
- **Random Forest:** 0.8923
- **KNN:** 0.8915
- **Decision Tree:** 0.8183

Observation

Linear, distance-based, and ensemble models show strong and comparable baseline performance. The single Decision Tree underperforms, highlighting the risk of overfitting and the benefit of ensembling.

Part 4: Baseline Evaluation on Hold-Out Test Set

"The ultimate test of generalization."

Evaluation Metrics Overview

Models were evaluated on the untouched 20% hold-out test set using:

- **Accuracy:** Overall correctness (use with caution in imbalanced data).
- **Precision:** Minimizing False Positives (unnecessary medical interventions).
- **Recall (Sensitivity):** Minimizing False Negatives (missed diagnoses). *Crucial for AHD.*
- **F1-Score:** Harmonic mean of Precision and Recall.
- **ROC-AUC:** Overall separability across all thresholds.
- **PR-AUC:** Performance specifically focused on the positive (minority) class.

Performance Metrics Comparison

Model	Acc	Prec	Rec	F1	ROC	PR
LR	0.885	0.839	0.929	0.881	0.952	0.919
KNN	0.902	0.893	0.893	0.893	0.969	0.957
SVM	0.902	0.867	0.929	0.897	0.955	0.922
DT	0.787	0.778	0.750	0.764	0.839	0.837
RF	0.918	0.871	0.964	0.915	0.962	0.952

- **Random Forest** achieves the highest Accuracy, Recall, and F1-Score.
- **KNN** achieves the highest ROC-AUC and PR-AUC.

Confusion Matrix: Random Forest

		Predicted	
		Negative	Positive
2*Actual	Negative	29 (TN)	4 (FP)
	Positive	1 (FN)	27 (TP)

Clinical Interpretation

- **False Negatives:** Only 1. The model missed just one patient with AHD.
- **Recall:** 0.964. Exceptional sensitivity, making it highly reliable for screening purposes where missing a disease is unacceptable.

Confusion Matrix: K-Nearest Neighbors

		Predicted	
		Negative	Positive
2*Actual	Negative	30 (TN)	3 (FP)
	Positive	3 (FN)	25 (TP)

Clinical Interpretation

- **Error Profile:** Very balanced (3 FP, 3 FN).
- **Overall Discrimination:** Despite a slightly lower recall than RF, its superior ROC-AUC (0.969) indicates it has the best overall ranking capability across all probability thresholds.

ROC Curve Analysis

- The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate against the False Positive Rate across all thresholds.
- **Results:**
 - KNN (0.969) and Random Forest (0.962) show near-perfect separation.
 - SVM (0.955) and Logistic Regression (0.952) also demonstrate excellent discriminative power.
 - Decision Tree (0.839) lags significantly behind, confirming that ensemble methods are necessary for this dataset.

Takeaway

All models except the single Decision Tree are highly capable of distinguishing between AHD and non-AHD patients.

Clinical Context: Precision vs. Recall

The Asymmetric Cost of Errors

In medical diagnostics, False Negatives and False Positives carry vastly different consequences.

- **Prioritizing Recall:** Missing a patient with AHD (False Negative) could lead to a fatal cardiac event. We must catch as many true cases as possible.
- **Accepting Lower Precision:** Flagging a healthy patient for an extra test (False Positive) causes financial cost and temporary anxiety, but is generally safe.

The Golden Rule

In this clinical context, **Random Forest** is the preferred model due to its maximized Recall (0.964), ensuring the highest safety net for patients.

Part 5: Feature Importance & Interpretation

"Extracting clinical insights from the black box."

Feature Importance: Random Forest

Extracted using **Gini Impurity** (Mean Decrease Impurity).

- **Thal:** 0.152
- **ChestPain:** 0.150
- **Ca:** 0.143
- **MaxHR:** 0.106
- **Oldpeak:** 0.090

Interpretation

Tree-based importance highlights features that provide the best hierarchical splits. Thalassemeia test results, chest pain type, and vessel coloring are the primary drivers of the model's decisions.

Feature Importance: K-Nearest Neighbors

Extracted using **Permutation Importance** (Model-agnostic).

- **Slope:** 0.079
- **Chol:** 0.044
- **Thal:** 0.039
- **RestECG:** 0.039
- **ExAng:** 0.035

Interpretation

Permutation importance measures how much the model's accuracy drops when a feature's values are randomly shuffled. Slope of the ST segment and Cholesterol levels have the highest global impact on KNN's distance calculations.

Comparing Feature Importance Methods

- **Gini Impurity (Tree-based):**

- Intrinsic to the model.
- Can be biased towards high-cardinality categorical features.
- Reflects internal decision-making logic.

- **Permutation Importance (Model-agnostic):**

- Evaluates impact on final predictions.
- Unbiased by feature type or scale.
- Reflects true global predictive value.

Best Practice

Use multiple methods to triangulate the most critical clinical predictors.

Top Predictors for AHD

Across both models and methods, certain clinical features consistently emerge as critical:

- **Thal (Thalassemia Test):** A strong indicator of underlying cardiovascular anomalies.
- **ChestPain:** Symptom presentation is a primary driver for clinical suspicion.
- **Ca (Major Vessels Colored):** Direct anatomical evidence of blockage.
- **MaxHR & Oldpeak:** Physiological stress response indicators during exercise.

These align perfectly with established cardiological risk factors, validating the model's clinical relevance.

- **K-Nearest Neighbors (KNN):**

- *Pros:* Highest overall discrimination (ROC-AUC 0.969), very balanced error profile.
- *Cons:* Computationally expensive during inference (calculating distances to all training points).

- **Random Forest (RF):**

- *Pros:* Highest Accuracy (0.918) and Recall (0.964), robust to overfitting, provides intrinsic feature importance.
- *Cons:* Slightly lower overall ROC-AUC compared to KNN.

Decision

For a screening tool where missing a case is critical, **Random Forest** is the optimal choice.

Summary of Findings

- ① **Robust Pipelines:** Embedding preprocessing in pipelines successfully prevented data leakage and handled missing values gracefully.
- ② **Ensemble Superiority:** Random Forest significantly outperformed the single Decision Tree, proving the value of bagging.
- ③ **Clinical Alignment:** By optimizing for Recall and utilizing Stratified CV, we ensured the model aligns with the real-world cost of medical errors.
- ④ **Actionable Insights:** Feature importance analysis confirmed that the model relies on medically sound predictors (Thal, ChestPain, Ca).

Best Practices for Clinical ML

- **Always Stratify:** Use stratified splits and cross-validation for imbalanced clinical data.
- **Look Beyond Accuracy:** Evaluate using ROC-AUC, PR-AUC, and Recall. Accuracy can be highly misleading.
- **Prevent Leakage:** Never scale or impute using the entire dataset before splitting. Use Pipeline and ColumnTransformer.
- **Context is King:** A model with 99% accuracy but 0% recall is useless in disease screening. Always define the clinical cost of errors.
- **Interpretability:** Use permutation importance to understand model behavior, especially for non-tree algorithms.

Questions?

Lumumba W. Victor

Trainer — Data Analyst — ML Expert

MediaCrest Training College